



**PEAK ACHIEVERS
ONLINE ACADEMY**

Learn • Practise • Improve • Achieve

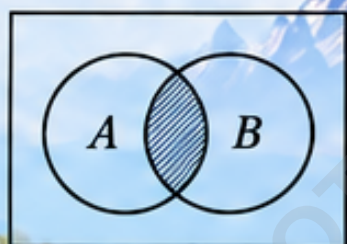
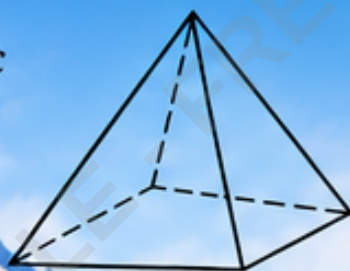
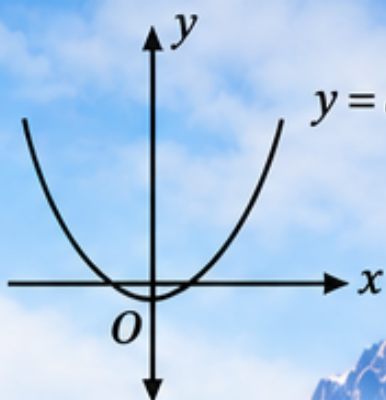
**GRADE
12**

CORE MATHEMATICS

REVISION BOOK

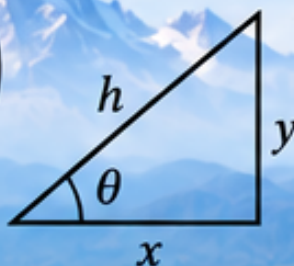
BOOK 2

PRACTICE • APPLICATIONS • EXAMINATION-STYLE TESTS



$$A = \begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$$

$$P = A(1 + r)^n$$



★
**UPDATED
FOR THE LATEST
CBE CURRICULUM**



BY
HEZEKIAH OMWERI NYATUKA



EDITOR & CONTENT VERIFIER

MR KIMANI PETER

Qualified Mathematics Teacher
Mount Kenya University

GRADE 12 CORE MATHEMATICS REVISION BOOK

BOOK 2 - PRACTICE, MASTERY & EXAMINATION

Author: Hezekiah Omweri Nyatuka

Mathematics Editor & Content Verifier

Mr Kimani Peter - Qualified Mathematics Teacher - Mount Kenya University

Peak Achievers Online Academy | Kenya | 2026

Website: peakachieversacademy.co.ke

Email: hezekiahomweri@gmail.com

Contacts: 0740166105 / 0740355410

Independent educational revision resource. No ISBN.

CHAPTER 1. MATRICES II

Order, determinant, singular matrices, inverse matrices, identity matrix and solving simultaneous equations using $AX=B$.

REMEMBER: Every worked solution below shows the mathematical path, not only the answer.

Practice Questions

1. Find $\det \begin{bmatrix} 7 & 3 \\ 2 & 5 \end{bmatrix}$.
2. Determine whether $\begin{bmatrix} 6 & 9 \\ 2 & 3 \end{bmatrix}$ is singular.
3. Find the inverse of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.
4. Solve $3x+2y=16$ and $x+2y=8$ using matrices.
5. Multiply $\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ by $\begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}$.
6. Find k if $\begin{bmatrix} k & 4 \\ 3 & 6 \end{bmatrix}$ is singular.

Fully Worked Step-by-Step Solutions

1. Find $\det \begin{bmatrix} 7 & 3 \\ 2 & 5 \end{bmatrix}$.

For $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\det A = ad - bc$.

$$\begin{aligned}\det A &= (7)(5) - (3)(2) \\ &= 35 - 6 \\ &= 29\end{aligned}$$

Final answer: $\det A = 29$.

2. Determine whether $\begin{bmatrix} 6 & 9 \\ 2 & 3 \end{bmatrix}$ is singular.

A matrix is singular when its determinant is zero.

$$\begin{aligned}\det A &= (6)(3) - (9)(2) \\ &= 18 - 18 = 0\end{aligned}$$

Therefore the matrix is singular and has no inverse.

3. Find the inverse of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

First find the determinant.

$$\det A = (4)(3) - (1)(2) = 12 - 2 = 10$$

Use $A^{-1} = 1/(ad-bc)\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

$$A^{-1} = 1/10 \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$$

Final answer: $A^{-1} = 1/10\begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$.

4. Solve $3x+2y=16$ and $x+2y=8$ using matrices.

Write $AX=B$.

$$\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 16 \\ 8 \end{bmatrix}$$

$$\det A = (3)(2) - (2)(1) = 4$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix}$$

$$x = (32 - 16) / 4 = 4$$

$$y = (-16 + 24) / 4 = 2$$

Final answer: $x=4$, $y=2$.

5. Multiply $\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ by $\begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix}$.

Multiply each row of the first matrix by each column of the second.

$$\begin{aligned} AB &= \begin{bmatrix} 2(1)+1(2) & 2(5)+1(3) \\ 3(1)+4(2) & 3(5)+4(3) \end{bmatrix} \\ &= \begin{bmatrix} 4 & 13 \\ 11 & 27 \end{bmatrix} \end{aligned}$$

Final answer: $\begin{bmatrix} 4 & 13 \\ 11 & 27 \end{bmatrix}$.

6. Find k if $\begin{bmatrix} k & 4 \\ 3 & 6 \end{bmatrix}$ is singular.

For a singular matrix, determinant=0.

$$6k - (4)(3) = 0$$

$$6k - 12 = 0$$

$$6k = 12$$

$$k = 2$$

Final answer: $k=2$.

CHECK YOUR WORK: Recalculate independently, keep exact values until the final step, and include units where required.

CHAPTER 2. VECTORS II

Vector components, magnitude, position vectors, parallelism, collinearity and internal division.

REMEMBER: Every worked solution below shows the mathematical path, not only the answer.

Practice Questions

1. A(-2,3), B(6,9): find AB.
2. Find $|(5,12)|$.
3. $a=(3,-2)$, $b=(5,4)$: find $2a-b$.
4. Show (8,12) and (2,3) are parallel.
5. Show A(1,2),B(4,8),C(7,14) are collinear.
6. P divides A(1,4),B(11,9) in ratio 2:3. Find P.

Fully Worked Step-by-Step Solutions

1. A(-2,3), B(6,9): find AB.

$$\begin{aligned} AB &= B-A = (6-(-2), 9-3) \\ &= (8,6) \end{aligned}$$

Final answer: $AB=(8,6)$.

2. Find $|(5,12)|$.

$$\begin{aligned} |v| &= \sqrt{5^2+12^2} \\ &= \sqrt{25+144} = \sqrt{169} \\ &= 13 \end{aligned}$$

Final answer: 13.

3. $a=(3,-2)$, $b=(5,4)$: find $2a-b$.

$$\begin{aligned} 2a &= (6,-4) \\ 2a-b &= (6,-4)-(5,4) \\ &= (1,-8) \end{aligned}$$

Final answer: (1,-8).

4. Show (8,12) and (2,3) are parallel.

$$(8,12)=4(2,3)$$

One vector is a scalar multiple of the other, so they are parallel.

5. Show A(1,2),B(4,8),C(7,14) are collinear.

$$\overrightarrow{AB} = (4-1, 8-2) = (3, 6)$$

$$\overrightarrow{BC} = (7-4, 14-8) = (3, 6)$$

$\overrightarrow{AB}$ and $\overrightarrow{BC}$ have the same direction, so A, B and C are collinear.

6. P divides A(1,4),B(11,9) in ratio 2:3. Find P.

For $AP:PB=2:3$, $P = (3A+2B)/5$.

$$P = (3(1,4)+2(11,9))/5$$

$$= ((3,12)+(22,18))/5$$

$$= (25,30)/5 = (5,6)$$

Final answer: $P=(5,6)$.

CHECK YOUR WORK: Recalculate independently, keep exact values until the final step, and include units where required.

CHAPTER 10. TRIGONOMETRY III

Equations, identities, ratios and periodic models.

REMEMBER: Every worked solution below shows the mathematical path, not only the answer.

Practice Questions

1. Solve $\sin \theta = \frac{\sqrt{3}}{2}$ for $0-360$.
2. Solve $\cos \theta = -\frac{1}{2}$ for $0-360$.
3. If $\sin \theta = \frac{5}{13}$ acute, find $\cos$ and $\tan$.
4. For $y = 5\sin 3x + 8$ find amplitude, period, maximum and minimum.

Fully Worked Step-by-Step Solutions

1. Solve $\sin \theta = \frac{\sqrt{3}}{2}$ for $0-360$.

Reference angle 60 degrees; sine positive in quadrants I and II.

$$\theta = 60 \text{ degrees, } 120 \text{ degrees}$$

2. Solve $\cos \theta = -\frac{1}{2}$ for $0-360$.

Reference angle 60 degrees; cosine negative in quadrants II and III.

$$\theta = 120 \text{ degrees, } 240 \text{ degrees}$$

3. If $\sin \theta = \frac{5}{13}$ acute, find $\cos$ and $\tan$.

$$\text{adjacent} = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$$

$$\cos \theta = \frac{12}{13}$$

$$\tan \theta = \frac{5}{12}$$

4. For $y = 5\sin 3x + 8$ find amplitude, period, maximum and minimum.

$$\text{Amplitude} = |5| = 5$$

$$\text{Period} = 360/3 = 120 \text{ degrees}$$

$$\text{Maximum} = 8 + 5 = 13$$

$$\text{Minimum} = 8 - 5 = 3$$

CHECK YOUR WORK: Recalculate independently, keep exact values until the final step, and include units where required.

CHAPTER 11. THREE-DIMENSIONAL GEOMETRY

Face diagonals, space diagonals, horizontal projections and elevation angles.

REMEMBER: Every worked solution below shows the mathematical path, not only the answer.

Practice Questions

1. Cuboid 9x12x20: find space diagonal.
2. Pole 12 m with cable 20 m: find horizontal distance and angle.
3. Find distance A(2,1,4) to B(8,5,7).

Fully Worked Step-by-Step Solutions

1. Cuboid 9x12x20: find space diagonal.

$$\begin{aligned} d &= \sqrt{9^2 + 12^2 + 20^2} \\ &= \sqrt{81 + 144 + 400} = \sqrt{625} = 25 \end{aligned}$$

Final answer: 25 units.

2. Pole 12 m with cable 20 m: find horizontal distance and angle.

$$\begin{aligned} \text{horizontal} &= \sqrt{20^2 - 12^2} \\ &= \sqrt{400 - 144} = \sqrt{256} = 16 \text{ m} \\ \tan \theta &= 12/16 = .75 \\ \theta &= \tan^{-1}(.75) = 36.9 \text{ degrees} \end{aligned}$$

3. Find distance A(2,1,4) to B(8,5,7).

$$\begin{aligned} d &= \sqrt{(8-2)^2 + (5-1)^2 + (7-4)^2} \\ &= \sqrt{36 + 16 + 9} = \sqrt{61} \\ &= 7.81 \end{aligned}$$

CHECK YOUR WORK: Recalculate independently, keep exact values until the final step, and include units where required.

MIXED CONCEPTS - GRADE 9 TO GRADE 12

Mixed Concepts 1 - Grade 9 Foundation

1. Solve $3x+7=25$.

$$3x+7=25$$

$$3x=25-7=18$$

$$x=18/3=6$$

2. Rectangle 12m by 8m: area and perimeter.

$$\text{Area}=lw=12(8)=96 \text{ m}^2$$

$$\text{Perimeter}=2(l+w)=2(12+8)=40 \text{ m}$$

3. Mean of 4,6,8,10,12.

$$\text{mean}=(4+6+8+10+12)/5=40/5=8$$

Mixed Concepts 2 - Grade 10 Bridge

1. Solve $x^2-7x+12=0$.

$$x^2-7x+12=(x-3)(x-4)$$

$$x=3 \text{ or } x=4$$

2. Simplify $\sqrt{72}$.

$$\sqrt{72}=\sqrt{36 \cdot 2}=6\sqrt{2}$$

3. Simple interest on KSh20,000 at 8% for 3 years.

$$I=Prt$$

$$=20000(.08)(3)$$

$$=\text{KSh}4,800$$

Mixed Concepts 3 - Grade 11 Bridge

1. Find $\det \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$.

$$\det=(4)(3)-(2)(1)=12-2=10$$

2. Sequence 5,8,11,... find 20th term.

$$a=5, d=3$$

$$T_n=a+(n-1)d$$

$$T_{20}=5+19(3)=62$$

3. $P(A)=.6, P(B)=.5, P(A \cap B)=.2$: union.

$$=.6+.5-.2=.9$$

FINAL NATIONAL EXAMINATION-STYLE PRACTICE TEST 1

Time: 2 Hours 30 Minutes | Total: 100 Marks

Fully Worked Marking Scheme - Calculation Standard**1. Find inverse of $\begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}$.**

$$\det = (5)(4) - (2)(3) = 20 - 6 = 14$$

$$A^{-1} = \frac{1}{14} \begin{bmatrix} 4 & -2 \\ -3 & 5 \end{bmatrix}$$

2. For $a = (3, -2)$, $b = (5, 4)$, find $2a - b$ and magnitude.

$$2a - b = (6, -4) - (5, 4) = (1, -8)$$

$$\text{magnitude} = \sqrt{1^2 + (-8)^2} = \sqrt{65} = 8.06$$

3. Evaluate integral 1 to 3 of $(3x^2 + 2x)$.

$$= [x^3 + x^2]_1^3$$

$$= (27 + 9) - (1 + 1) = 34$$

4. KSh80,000 at 7.5% compound for 3 years.

$$A = 80000(1.075)^3$$

$$= 80000(1.242296875) = \text{KSh}99,383.75$$

$$\text{Interest} = 99,383.75 - 80,000 = \text{KSh}19,383.75$$

5. LP: $x + y \leq 30$, $2x + y \leq 40$; $P = 600x + 900y$.Intersection: subtract equations $\rightarrow x = 10, y = 20$ Vertices: $(0, 0), (20, 0), (10, 20), (0, 30)$.

Profits: 0; 12,000; 24,000; 27,000

Maximum = KSh27,000 at $(0, 30)$.

MARKING NOTE: Method marks should reward the correct rule/formula and substitution. Arithmetic and final-answer marks follow. Units and interpretation are required where the context demands them.

FINAL NATIONAL EXAMINATION-STYLE PRACTICE TEST 6

Time: 2 Hours 30 Minutes | Total: 100 Marks

Fully Worked Marking Scheme - Calculation Standard**1. Commerce: $5x+3y=6000, 3x+4y=5700$.**

$$A = \begin{bmatrix} 5 & 3 \\ 3 & 4 \end{bmatrix}, \det = 20 - 9 = 11$$

$$A^{-1} = \frac{1}{11} \begin{bmatrix} 4 & -3 \\ -3 & 5 \end{bmatrix}$$

$$x = (4(6000) - 3(5700)) / 11 = 6900 / 11 = \text{KSh}627.27$$

$$y = (-3(6000) + 5(5700)) / 11 = 10500 / 11 = \text{KSh}954.55$$

2. LP multiple optimum: $x+y \leq 24, 2x+y \leq 36; P=1600x+800y$.

Vertices: (0,0), (18,0), (12,12), (0,24).

$$P(18,0) = 28,800$$

$$P(12,12) = 19,200 + 9,600 = 28,800$$

$$P(0,24) = 19,200$$

$$P = 800(2x+y)$$

Therefore every feasible point on $2x+y=36$ between (18,0) and (12,12) gives the maximum KSh28,800.**3. Cost $C=2x^2-120x+5000$.**

$$C' = 4x - 120$$

$$4x - 120 = 0 \rightarrow x = 30$$

$$C'' = 4 > 0, \text{ so minimum}$$

$$C(30) = 2(900) - 3600 + 5000 = 3200$$

Minimum modelled cost is KSh3,200 at 30 units.

4. Finance 600,000 at 10% 4y vs equipment 800,000 inflation 5% 4y.

$$\text{Investment} = 600000(1.10)^4 = 600000(1.4641) = \text{KSh}878,460$$

$$\text{Equipment} = 800000(1.05)^4 = 800000(1.21550625) = \text{KSh}972,405$$

$$\text{Shortfall} = 972405 - 878460 = \text{KSh}93,945$$

MARKING NOTE: Method marks should reward the correct rule/formula and substitution. Arithmetic and final-answer marks follow. Units and interpretation are required where the context demands them.

ABOUT THIS BOOK

This is Book 2 of the Peak Achievers Grade 12 Core Mathematics Revision Series. It is designed to give learners extensive practice, real-life applications and examination-style questions with step-by-step solutions. The book covers the entire Grade 12 curriculum in a clear, structured and learner-friendly way, helping you build confidence and mastery in Mathematics.

KEY FEATURES

- ✓ Complete coverage of the Grade 12 Core Mathematics syllabus
- ✓ Topic practice questions with step-by-step solutions
- ✓ Real-life and examination-style application questions
- ✓ Grade 9 – 12 mixed concept questions for deeper understanding
- ✓ Six 100-mark National Examination–Style Practice Tests
- ✓ Full worked marking schemes with detailed steps
- ✓ Diagrams, graphs and mathematical illustrations
- ✓ Final revision checklist

IDEAL FOR

- ✓ Grade 12 learners
- ✓ Teachers and tutors
- ✓ Revision and exam preparation
- ✓ Self-study and extra practice

OTHER BOOKS IN THE SERIES

- ✓ Book 1: Complete Notes & Worked Examples
- ✓ Book 2: Practice, Applications & Examination-Style Tests

*Mathematics Today
A Brighter Tomorrow*



**PEAK ACHIEVERS
ONLINE ACADEMY**

Learn • Practise • Improve • Achieve



www.peakachieversacademy.co.ke



hezekiaomweri@gmail.com



0740166105 / 0740355410



Peak Achievers Online Academy